
Technical University of Crete
School of Electrical and Computer Engineering
Course: **Optimization**
Exercise 1 (80/1000)
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In this exercise, we will study simple concepts from Calculus of Several Variables (Taylor expansions), Convex Sets, and Convex Functions (Note: $\mathbb{R}_+ = \{x \in \mathbb{R} \mid x \geq 0\}$, $\mathbb{R}_{++} = \{x \in \mathbb{R} \mid x > 0\}$).

You must prepare an electronic (LaTeX) report
and upload the pdf file and the related code.

1. (10) Let $f : \mathbb{R}_+ \rightarrow \mathbb{R}$, with $f(x) = \frac{1}{1+x}$. Let $x_0 \in \mathbb{R}_+$, and define the first- and second-order Taylor approximations of f at x_0 as

$$\begin{aligned} f_{(1)}(x) &= f(x_0) + f'(x_0)(x - x_0), \\ f_{(2)}(x) &= f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2. \end{aligned} \tag{1}$$

- (a) (5) Find the analytic expressions for functions f' and f'' ;

By differentiating, we acquire the first and second derivatives of f as:

$$\begin{aligned} f'(x) &= -\frac{1}{(x+1)^2}, \\ f''(x) &= \frac{d}{dx} \left[-\frac{1}{(x+1)^2} \right] \stackrel{u=(x+1)}{=} -\frac{d}{du} \left(\frac{1}{u^2} \right) \cdot \frac{du}{dx} = -\left(\frac{-1}{u^3} \right) \cdot 1 \stackrel{u=(x+1)}{=} \frac{1}{(x+1)^3} \end{aligned}$$

- (b) (5) Draw in a common plot $f(x)$, $f_{(1)}(x)$ and $f_{(2)}(x)$ and, in order to understand the behavior of the approximations, consider various values of x_0 .

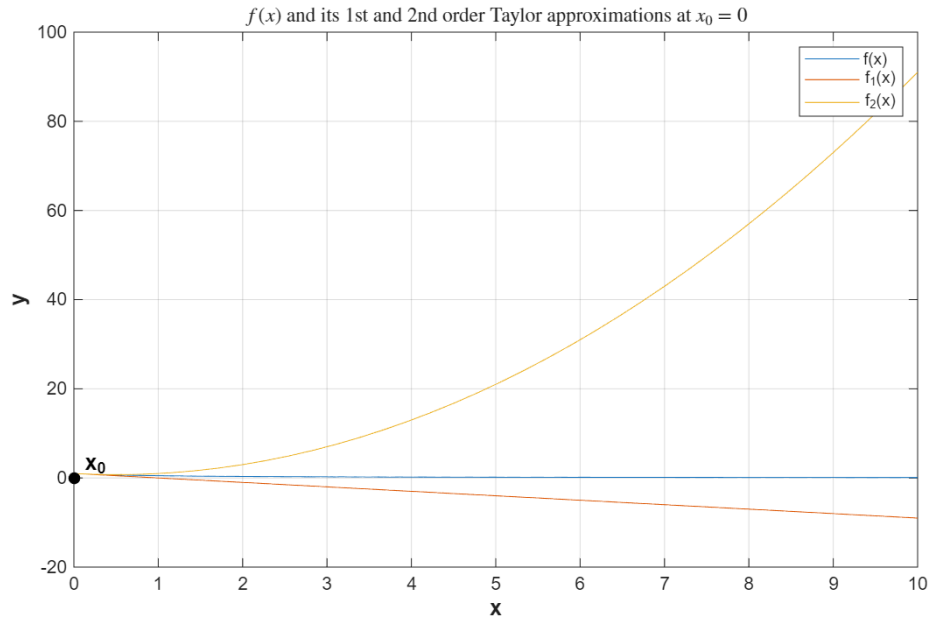


Figure 1: f and its first and second order Taylor approximations, at $x_0 = 0$

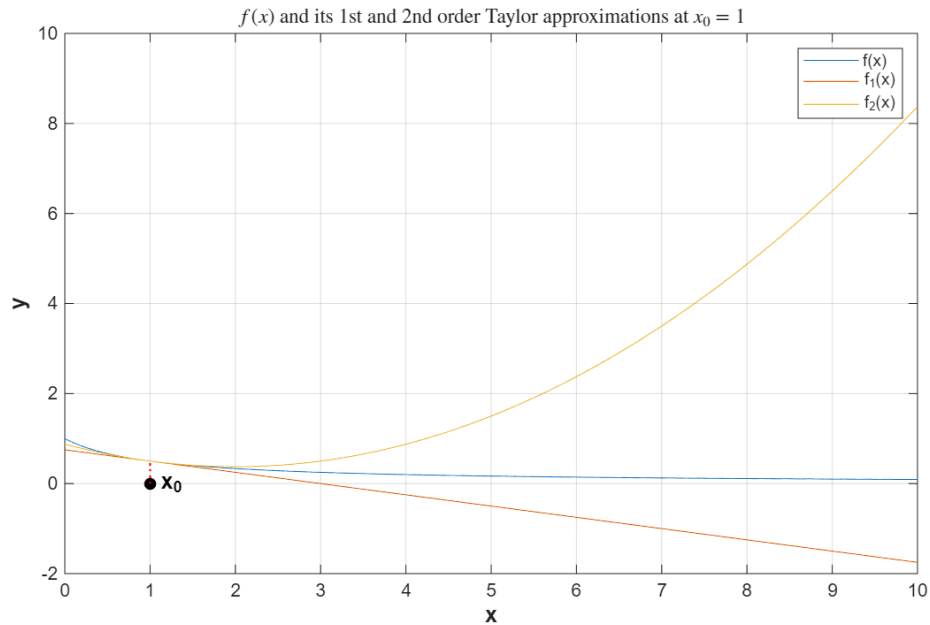


Figure 2: f and its first and second order Taylor approximations, at $x_0 = 1$

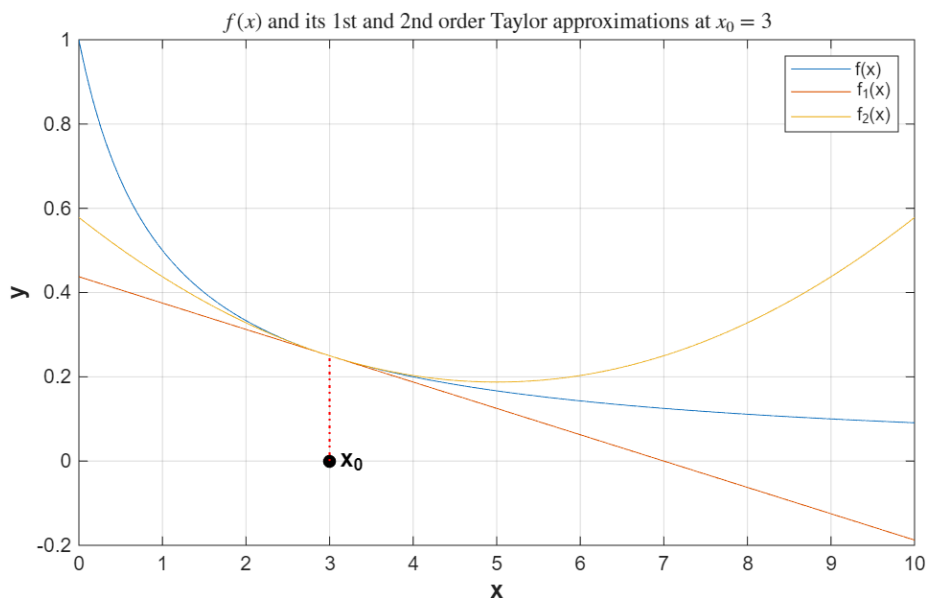


Figure 3: f and its first and second order Taylor approximations, at $x_0 = 3$

As we can see, the different approximations as given by the Taylor expansions of f , give a somewhat accurate picture of f , up to a certain small range around the point of approximation x_0 .

Of course, if we were to calculate higher-order approximations (larger than second order), we would get curves that would fit for a bigger range of x values with the curve of f .

2. (20) Let $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$.

(a) (0) Compute and plot, via `mesh`, f for $x_1, x_2 \in [0, x_*]$, with $x_* > 0$.

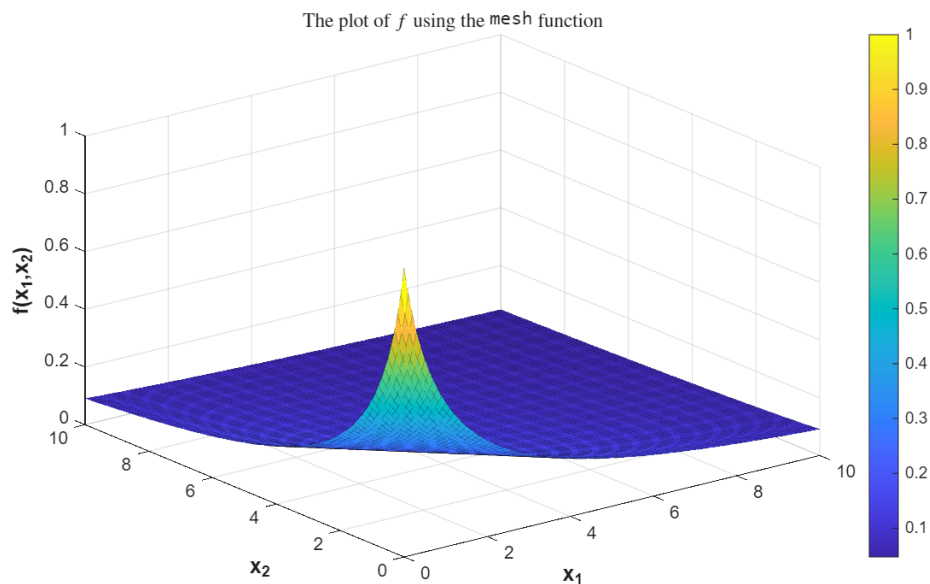


Figure 4: f in the $[0, x_*] \times [0, x_*]$ region, using the `mesh` MatLab function

(b) (5) Plot the level sets of f , via `contour`. What do you observe? Can you explain the phenomenon?

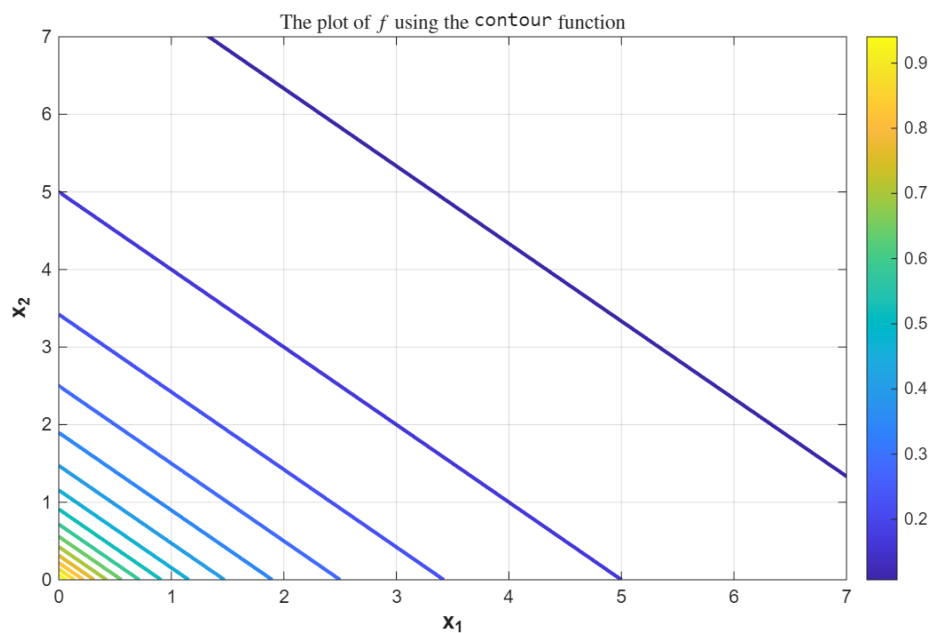


Figure 5: f in the $[0, x_*] \times [0, x_*]$ region, using the `contour` MatLab function

Firstly, let's clarify that the point $\mathbf{0} = [0 \ 0]$ is the maximum of the function f (in the domain $[0, x_*] \times [0, x_*]$), something that can be verified by both the `mesh` plot in (a), as well as its formula; the fraction $\frac{1}{1+x_1+x_2}$, can attain a value of up to 1, for positive (or zero) values of x_1, x_2 .

We can expect that f , will have a parabola-like behavior similar to $\frac{1}{x}$, in the 1D case. This means that it will start getting larger and larger values, with an increasing rate, as we approach the maximum point of its domain.

Indeed, as we can see the level sets of the function f get denser and denser, implying that f changes with an increasing rate.

Also, the colorbar helps us realize that as the level-sets get denser, the values of f increase, meaning that the rate of change on points of the different level-sets is actually positive.

All in all, the increasing density of the level-sets as well as the brightening on their color-code help us understand that the values of f increase with an increasing rate, as we approach its maximum point.

- (c) (5) Compute the first- and second-order Taylor approximations of f at point $\mathbf{x}_0 = (x_{0,1}, x_{0,2})$.

We choose our point of approximation $\mathbf{x}_0$ to be $\mathbf{x}_0 = [0 \ 0]^T$.

The 1st order Taylor approximation of f at the point $\mathbf{x}_0$ is given by $f_1(\mathbf{x}_0) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0)$. So we have to calculate the gradient of the function of f

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{1}{1+x_1+x_2} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{1}{1+x_1+x_2} \right) \end{bmatrix} = \begin{bmatrix} -\frac{1}{(1+x_1+x_2)^2} \\ -\frac{1}{(1+x_1+x_2)^2} \end{bmatrix}$$

which at point $\mathbf{x}_0 = [0 \ 0]^T$, acquires the value

$$\nabla f(\mathbf{x}_0) = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

giving us the first order Taylor approximation to be

$$f_1(\mathbf{x}_0) = \frac{1}{0+0+1} - \begin{bmatrix} 1 & 1 \end{bmatrix} \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) = 1 - \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1 - x_1 - x_2$$

The 2nd order Taylor approximation of f at the point $\mathbf{x}_0$ is given by $f_2(\mathbf{x}_0) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \nabla^2 f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0)$ which can also be written as $f_2(\mathbf{x}_0) = f_1(\mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T \nabla^2 f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0)$. Thus we have to calculate the hessian of f

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial^2 x_2} \end{bmatrix} = \begin{bmatrix} \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \\ \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \end{bmatrix}$$

which at point $\mathbf{x}_0 = [0 \ 0]^T$, acquires the value

$$\nabla^2 f(\mathbf{x}_0) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

giving us the second order Taylor approximation to be

$$\begin{aligned} f_2(\mathbf{x}_0) &= 1 - x_1 - x_2 + \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1 - x_1 - x_2 + \frac{1}{2} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 + x_2 \\ x_2 + x_2 \end{bmatrix} \\ &= 1 - x_1 - x_2 + x_1(x_1 + x_2) + x_2(x_1 + x_2) = 1 - x_1 - x_2 + (x_1 + x_2)^2 \end{aligned}$$

(d) (5) Draw on a common plot, via `mesh`, f and its first-order Taylor approximation.

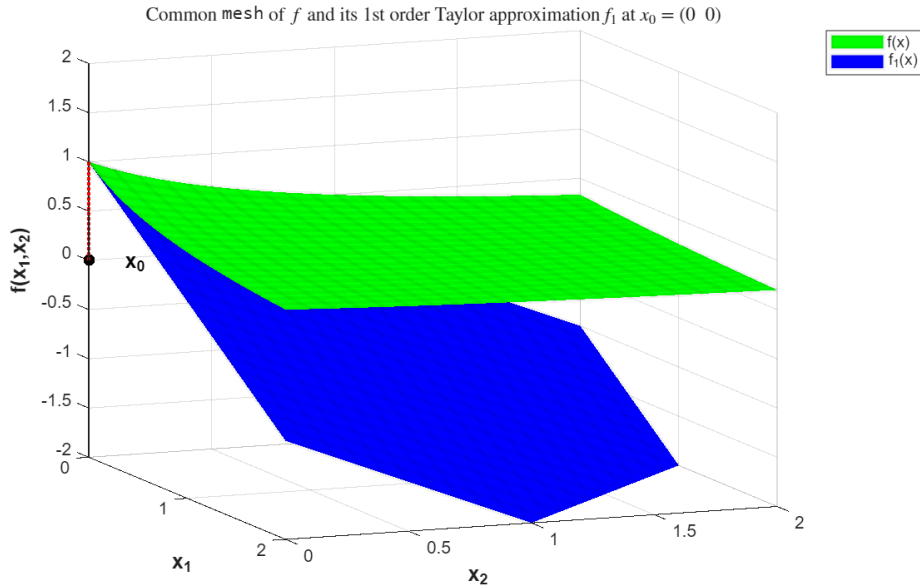


Figure 6: f and its first order Taylor approximation, at $x_0 = (0 \ 0)$, using the `mesh` MatLab function

- (e) (5) Draw on a common plot, via `mesh`, f and its second-order Taylor approxi-
Common `mesh` of f and its 2nd order Taylor approximation f_2 at $x_0 = (0 \ 0)$

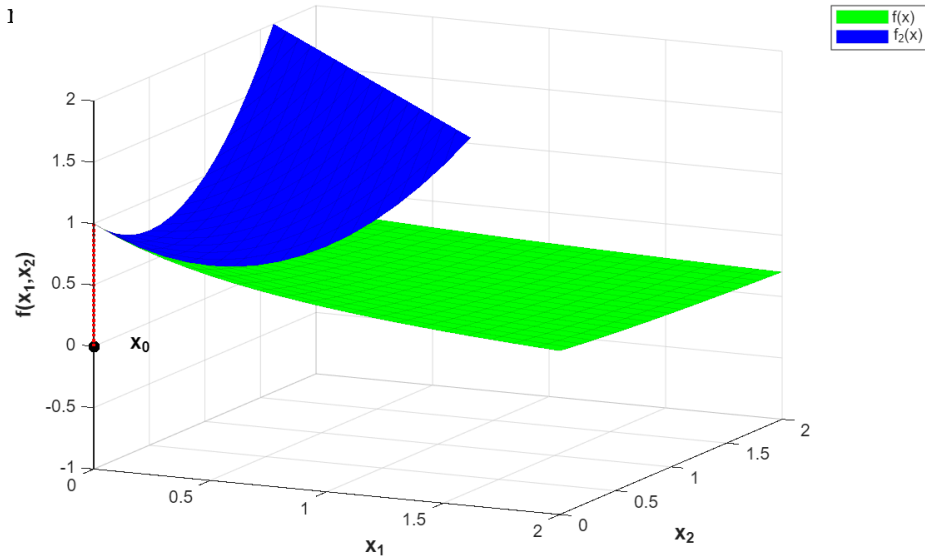


Figure 7: f and its second order Taylor approximations, at $x_0 = (0 \ 0)$, using the `mesh` MatLab function

As in the 1D case, in exercise 1, we can see that the approximations of f give a more precise picture as the order of the Taylor approximation increases.

3. (20) Let $\mathbb{S}_{\mathbf{a},b} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T \mathbf{x} \leq b\}$.

- (a) (5) Prove that $\mathbb{S}_{\mathbf{a},b}$ is convex.

Suppose $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{S}_{\mathbf{a},b}$ and $0 \leq \theta \leq 1$.

Then $\mathbf{a}^T \mathbf{x}_1 \leq b$ and $\mathbf{a}^T \mathbf{x}_2 \leq b$

$$\stackrel{\theta \geq 0}{\implies} \theta \mathbf{a}^T \mathbf{x}_1 \leq b \quad (1) \quad \text{and} \quad \stackrel{(1-\theta) \geq 0}{\implies} (1-\theta) \mathbf{a}^T \mathbf{x}_2 \leq (1-\theta)b \quad (2)$$

$$\begin{aligned} (1) + (2) &\Rightarrow \theta \mathbf{a}^T \mathbf{x}_1 + (1-\theta) \mathbf{a}^T \mathbf{x}_2 \leq \theta b + (1-\theta)b \\ &\Rightarrow \theta \mathbf{a}^T \mathbf{x}_1 + \mathbf{a}^T \mathbf{x}_2 - \theta \mathbf{a}^T \mathbf{x}_2 \leq \theta b + b - \theta b \\ &\Rightarrow \mathbf{a}^T [\underbrace{\theta \mathbf{x}_1 + (1-\theta) \mathbf{x}_2}_{\mathbf{x}_*}] \leq b \\ &\Rightarrow \mathbf{a}^T \mathbf{x}_* \leq b \end{aligned}$$

This of course means that $\mathbf{x}_* = \theta \mathbf{x}_1 + (1-\theta) \mathbf{x}_2 \in \mathbb{S}_{\mathbf{a},b}$, implying that every convex combination of elements inside of the set $\mathbb{S}_{\mathbf{a},b}$, are also elements of $\mathbb{S}_{\mathbf{a},b}$, thus making it a **convex set**.

(b) (15) Prove that $\mathbb{S}_{\mathbf{a},b}$ is *not* affine (a counterexample is sufficient).

Let's assume that $\mathbb{S}_{\mathbf{a},b}$ is indeed affine. Then, every pair of points in $\mathbb{R}^n$ will construct an affine combination $\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2$ that will also be in $\mathbb{S}_{\mathbf{a},b}$, for every real valued θ . For $n = 2$, we can assume $\mathbf{a} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$ and $b = 12$, as well as

the two vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ both of which are elements of $\mathbb{S}_{\mathbf{a},b}$ since

$$\mathbf{a}^T \mathbf{x}_1 = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 0 \cdot 1 + 2 \cdot 2 = 4 \leq b = 12$$

$$\mathbf{a}^T \mathbf{x}_2 = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = 0 \cdot 1 + 2 \cdot (-2) = -4 \leq b = 12$$

However, when checking whether their affine combination is an element of $\mathbb{S}_{\mathbf{a},b}$ we get the following:

$$\begin{aligned} \mathbf{a}^T [\theta \mathbf{x}_1 + (1 - \theta) \mathbf{x}_2] &\leq b \\ \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} \theta \\ 2\theta \end{bmatrix} + \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 - \theta \\ -2(1 - \theta) \end{bmatrix} &\leq 12 \\ 4\theta - 4(1 - \theta) &\leq 12 \\ 8\theta &\leq 8 \\ \theta &\leq 1 \end{aligned}$$

This means that, there is a constraint for the values of θ that the above inequality holds, leading us to a contradiction.

Thus, the initial assumption that any affine combination ($\forall \theta \in \mathbb{R}$) of two elements in $\mathbb{S}_{\mathbf{a},b}$ is also an element of $\mathbb{S}_{\mathbf{a},b}$ is wrong, making $\mathbb{S}_{\mathbf{a},b}$ **not** affine.

4. (10) Find the point $\mathbf{x}_*$ that is co-linear with $\mathbf{a}$ and lies on the hyperplane $\mathbb{H}_{\mathbf{a},b} := \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{a}^T \mathbf{x} = b\}$.

For two vectors $\mathbf{x}_*$, $\mathbf{a}$ ($\neq \mathbf{0}$) to be *co-linear* the following has to be true

$$\mathbf{x}_* = \lambda \mathbf{a}, \lambda \in \mathbb{R}$$

and since $\mathbf{x}_*$ lies on $\mathbb{H}_{\mathbf{a},b}$:

$$\begin{aligned} \mathbf{a}^T \mathbf{x}_* &= b \\ \mathbf{a} \mathbf{a}^T \mathbf{x}_* &= \mathbf{a} b \\ \|\mathbf{a}\|^2 \mathbf{x}_* &= b \mathbf{a} \\ \boxed{\mathbf{x}_* = \frac{b}{\|\mathbf{a}\|^2} \mathbf{a}} \end{aligned}$$

which is of the form $\mathbf{x}_* = \lambda \mathbf{a}$, with $\lambda = \frac{b}{\|\mathbf{a}\|^2}$

5. (20) Check whether the following functions are convex or not (using, for example, the second derivative rule).

(a) (5) $f : \mathbb{R}_+ \rightarrow \mathbb{R}$, with $f(x) = \frac{1}{1+x}$;

By differentiating, we acquire the first and second derivatives of f as:

$$f'(x) = -\frac{1}{(x+1)^2},$$

$$f''(x) = \frac{d}{dx} \left[-\frac{1}{(x+1)^2} \right] = \frac{1}{(x+1)^3} > 0 \quad \forall x \in \mathbb{R}_+$$

Thus, using the second-order condition we ensure that f is **convex**.

(b) (5) $f : \mathbb{R}_+^2 \rightarrow \mathbb{R}$, with $f(x_1, x_2) = \frac{1}{1+x_1+x_2}$;

In order to get the first derivative of f , that is, its **gradient**, we have to differentiate f with respect to all of its variables. Thus:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{1}{1+x_1+x_2} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{1}{1+x_1+x_2} \right) \end{bmatrix} = \begin{bmatrix} -\frac{1}{(1+x_1+x_2)^2} \\ -\frac{1}{(1+x_1+x_2)^2} \end{bmatrix}$$

By applying the same logic, we acquire the **Hessian** matrix of f :

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f(\mathbf{x})}{\partial x_1^2} & \frac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \frac{\partial^2 f(\mathbf{x})}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \\ \frac{2}{(1+x_1+x_2)^3} & \frac{2}{(1+x_1+x_2)^3} \end{bmatrix} \succeq \mathbf{O} \quad \forall x_1, x_2 \in \mathbb{R}_+$$

Thus, using the second-order condition we ensure that f is **convex**.

- (c) (5) $f : \mathbb{R}_{++} \rightarrow \mathbb{R}$, with $f(x) = x^a$, for (to get a better feeling, plot function x^a , for various values of a)

i. $a \geq 1$ and $a \leq 0$;

ii. $0 \leq a \leq 1$.

When differentiating with respect to x in x^a , the result we derive is the same in any case of range of values that the exercise defines for a , and writes as:

$$f' = \frac{d}{dx} x^a = ax^{(a-1)}$$

When differentiating again:

$$f'' = \frac{d}{dx} ax^{(a-1)} = a(a-1)x^{(a-2)}$$

Since the quantity $x^{(a-2)}$ is positive for any value of x in $\mathbb{R}_{++}$ (and any $a \in \mathbb{R}$), the sign of f'' has only to do with what sign the factor $a(a-1)$ has, in each case of study.

i. $a \geq 1$ and $a \leq 0$

- For $a \geq 1 > 0 \Rightarrow a(a-1) > 0$ making $f'' > 0 \forall x \in \mathbb{R}_{++}$ and f concave.

This can easily be confirmed visually, from the graphs below.

- For $a \leq 0 < 1 \Rightarrow a < 0$ and $(a-1) < 0 \Rightarrow a(a-1) > 0$ making $f'' > 0 \forall x \in \mathbb{R}_{++}$ and f concave.

This can easily be confirmed visually, from the graphs below.

ii. $0 \leq a \leq 1$

For $0 \leq a \leq 1$, while the a factor is positive, the factor $a-1$ factor is negative making their in between product $a(a-1)$ also negative. This means that in such a case f will be concave, something that can be confirmed visually from the following graphs.

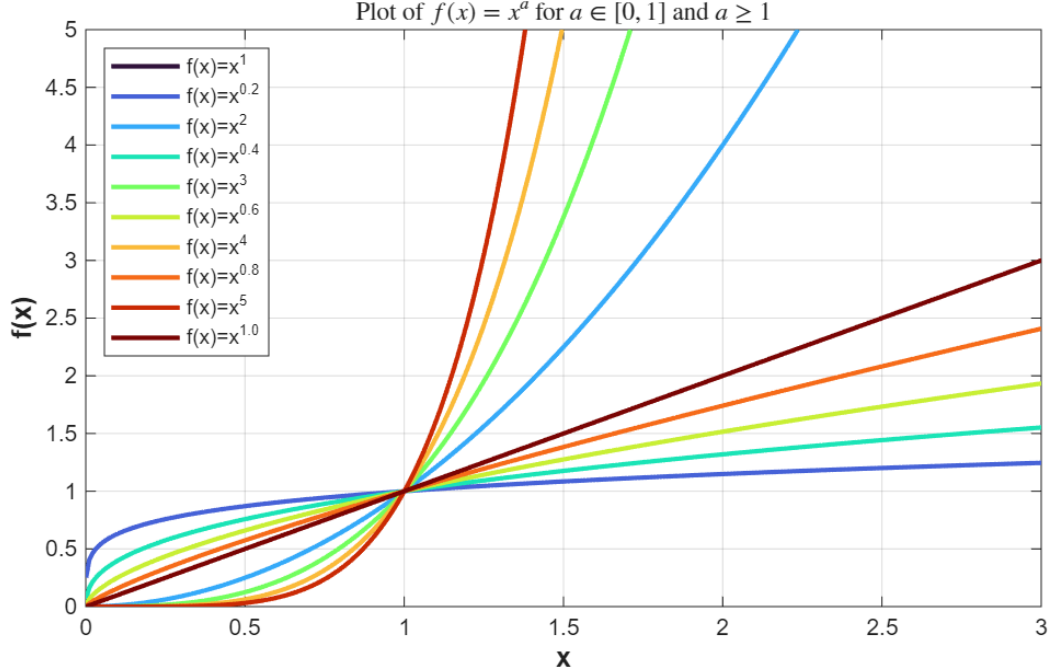


Figure 8: Common plot for $f(x) = x^a$ and different values of a . The versions of f with $a \geq 1$ have a clearly **convex** behavior, while the ones with $a \in [0, 1]$ have **concave** behavior

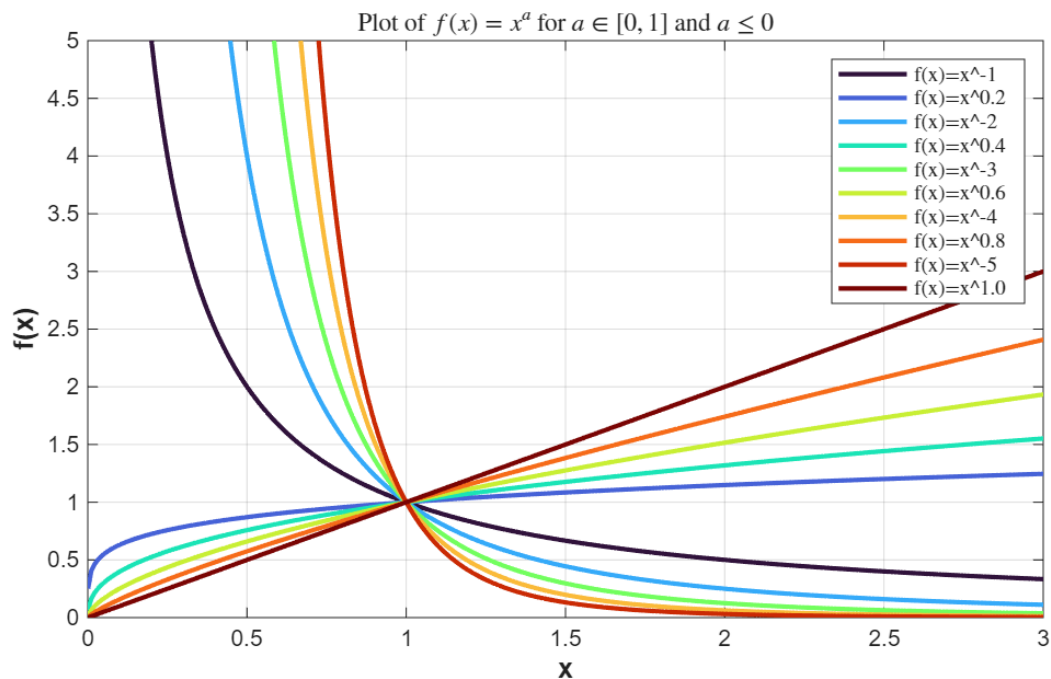


Figure 9: Common plot for $f(x) = x^a$ and different values of a . The versions of f with $a \leq 0$ have a clearly **convex** behavior, while the ones with $a \in [0, 1]$ have **concave** behavior

Conclusively, what we can say when studying the graphs, is that the convex behavior of the function x^a "decreases" as a decreases (while still $a > 1$), becomes concave for $a \in [0, 1]$ and then returns back to being convex for $a \leq 0$.

- (d) (5) $f_1, f_2 : \mathbb{R}^n \rightarrow \mathbb{R}$, with $f_1(\mathbf{x}) = \|\mathbf{x}\|_2$ and $f_2(\mathbf{x}) = \|\mathbf{x}\|_2^2$ (plot the functions for $n = 2$).

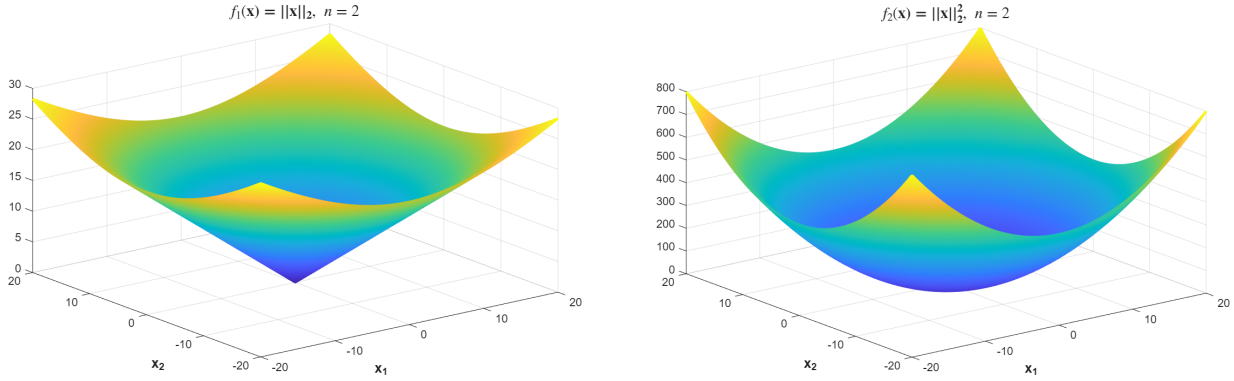


Figure 10: $f_1(\mathbf{x}) = \|\mathbf{x}\|_2$ (left) and $f_2(\mathbf{x}) = \|\mathbf{x}\|_2^2$ (right), $\forall \mathbf{x} \in \mathbb{R}^2$

In both cases, the graphs drawn just above reveal that f_1, f_2 are both convex functions.

This is something to be expected since, by applying the first and second order criterion to f_1 and f_2 respectively we get:

$$\begin{aligned} f_1(\theta \mathbf{x} + (1 - \theta) \mathbf{y}) &= \|\theta \mathbf{x} + (1 - \theta) \mathbf{y}\| \stackrel{(*)}{\leq} \|\theta \mathbf{x}\| + \|(1 - \theta) \mathbf{y}\| \\ &= \theta \|\mathbf{x}\| + (1 - \theta) \|\mathbf{y}\| = \theta f_1(\mathbf{x}) + (1 - \theta) f_1(\mathbf{y}) \end{aligned}$$

where at point (*) we used the triangle inequality.

$$\nabla f_2(\mathbf{x}) = \nabla \|\mathbf{x}\|^2 = \nabla (\mathbf{x}^T \mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} (x_1^2 + x_2^2) \\ \frac{\partial}{\partial x_2} (x_1^2 + x_2^2) \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} = 2\mathbf{x}$$

$$\nabla^2 f_2(\mathbf{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_1} & \frac{\partial}{\partial x_2} \frac{\partial f}{\partial x_1} \\ \frac{\partial}{\partial x_1} \frac{\partial f}{\partial x_2} & \frac{\partial}{\partial x_2} \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2\mathbb{I}_{n \times n} \succ 0$$

6. (10) In the sequel, we shall study the very important convex quadratic functions. Let $\mathbf{P} \in \mathbb{S}_{++}^2$, where $\mathbb{S}_{++}^2$ is the set of the (2×2) positive definite matrices, $\mathbf{q} \in \mathbb{R}^2$, and $r \in \mathbb{R}$.¹ Consider the convex quadratic function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined as:

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{P} \mathbf{x} + \mathbf{q}^T \mathbf{x} + r. \quad (3)$$

It can be shown (we shall prove it later) that the function f is minimized at point

$$\mathbf{x}_* = -\mathbf{P}^{-1} \mathbf{q}. \quad (4)$$

- (a) (10) Plot function f , via `mesh` and `contour` around $\mathbf{x}_*$, and verify, visually, the claim about the optimal point $\mathbf{x}_*$.

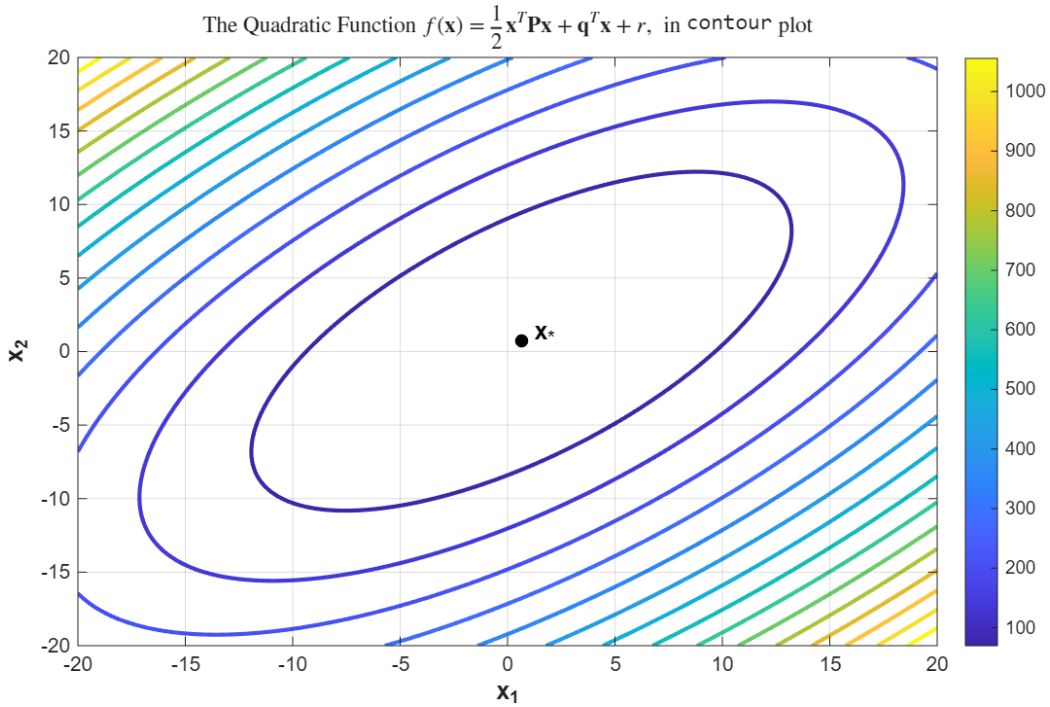


Figure 11: f plotted via `contour`, around the minimum point $\mathbf{x}_* = -\mathbf{P}^{-1} \mathbf{q}$

¹In order to generate such a $\mathbf{P}$, do as follows:

$$\begin{aligned} \mathbf{A} &= \text{randn}(2, 2); \\ \mathbf{P} &= \mathbf{A} * \mathbf{A}'; \end{aligned} \quad (2)$$

The generated matrix $\mathbf{P}$ is positive definite with probability 1.

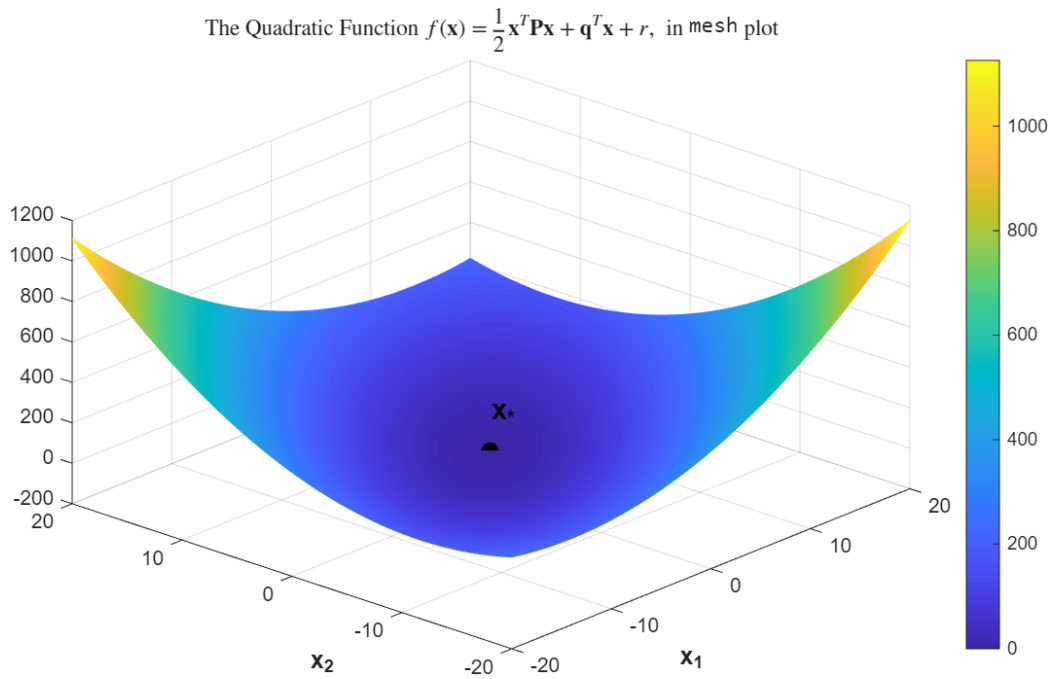


Figure 12: f plotted via `mesh`, around the minimum point $\mathbf{x}_* = -\mathbf{P}^{-1}\mathbf{q}$

After calculating the point $\mathbf{x}_*$ from the formula in the exercise description as $\mathbf{x}_* = -\mathbf{P}^{-1}\mathbf{q}$, we plot it in the same plot as f . By noticing the color interpretation in both the level set graph and the 3D graph, we can easily verify that the point $\mathbf{x}_*$ is indeed the minimum point of the function f .